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Efficient Algorithm for Identifying Repeating Patterns Modulo 4 and its Application in Complex Numbers

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Abstract

In this work, we present a modular arithmetic based algorithm to identify an object in a repeated infinite/finite sequence of 4 objects. We will theoretically justify the algorithm and present some of its applications, such as effectively computing the higher powers of complex number $\mathrm{i} = \sqrt{-1}$.

Keywords: Periodic Sequences, Analytical Methods, Modular Arithmetic, Number Classification, Number Classification, Even-odd Analysis, Computational Mathematics, Sequence Patterns, Exponentiation, Complex Numbers
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Number Classification; Even-odd Analysis; Computational Mathematics; Sequence Patterns; Exponentiation; Complex Numbers

1. Introduction

Periodic sequences, characterized by their repetitive patterns, find applications across various mathematical and computational domains. This paper presents a novel analytical approach for swiftly determining the position of a given number within such sequences, leveraging mathematical principles and modular arithmetic. Our methodology, founded upon the periodicity of sequences, classifies numbers into even and odd categories and subsequently employs distinct procedures for each classification.

For even numbers, a straightforward division by 2 method is applied, leading to the identification of the corresponding term within the sequence. Odd numbers, on the other hand, undergo an intricate transformation, involving a subtraction operation, followed by a similar division by 2. These processes yield unequivocal results and align with a fundamental numerical intuition that associates even exponents with positive terms and odd exponents with negative ones.

Illustrative examples and mathematical rigor substantiate the efficacy of our approach in rapidly and accurately identifying terms within periodic sequences. This analytical methodology not only simplifies complex calculations but also contributes to the understanding of the inherent properties of periodic sequences, laying the groundwork for diverse applications in mathematics and computational sciences.Periodic sequences, characterized by their repetitive patterns, find applications across various mathematical and computational domains. This paper presents a novel analytical approach for swiftly determining the position of a given number within such sequences, leveraging mathematical principles and modular arithmetic. Our methodology, founded upon the periodicity of sequences, classifies numbers into even and odd categories and subsequently employs distinct procedures for each classification.

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Illustrative examples and mathematical rigor substantiate the efficacy of our approach in rapidly and accurately identifying terms within periodic sequences. This analytical methodology not only simplifies complex calculations but also contributes to the understanding of the inherent properties of periodic sequences, laying the groundwork for diverse applications in mathematics and computational sciences. Given a periodic sequence S with a repeating pattern of length P, represented as

$$S = (a, b, c, d, \ldots, P)$$

where $(a, b, c, d, \ldots)$ denote distinct terms in the sequence, this study aims to devise analytical methodologies for the rapid determination of a term based on an integer N.

The periodic nature of S necessitates a cyclical analysis employing modular arithmetic. We endeavor to exploit this property by categorizing N into even (N_{even}) and odd (N_{odd}) classes. Each classification leads to a unique analytical procedure for swift determination of the corresponding term within the sequence.

By establishing a precise analytical framework, we aim to augment the efficiency and accuracy of identifying terms in periodic sequences, offering new avenues for computational expedience and mathematical comprehension.

2. Materials and Methods

In this section, we detail the materials, techniques, and algorithms employed to analyze periodic sequences and identify the position of given numbers within these sequences. Our methodology is anchored in the profound principles of modular arithmetic, a field with pivotal importance in number theory and mathematical cryptography.

2.1. Sequence Pattern Description

Our investigation delves into characterizing periodic sequences, denoted as

$$S = \{a, b, c, d, a, b, c, d, \ldots\}.$$

This showcases a cyclic behavior with a fixed period N, where N is a positive integer. This periodicity is fundamental, implying a reset of S every N term, forming the basis of our analysis.

Let us denote modulo operations:

$$\begin{aligned} 1 \mod 4 &\equiv 1 \implies a \\ 2 \mod 4 &\equiv 2 \implies b \\ 3 \mod 4 &\equiv 3 \implies c \\ 4 \mod 4 &\equiv 0 \implies d \end{aligned}$$

In this cyclical behavior, i repeats every four powers, i.e., $i^{4n} = d$, where $n \in \mathbb{Z}$, due to the remainder of modulo 4 being 0. We illustrate this with a large exponent, $N = 892713821789371238$.

Proving that only the last two digits are required to find the value of the complex number:

2.1.1. Exponent Partitioning

We split $N = 892713821789371238$ into two components: 892713821789371200 and 38.

		Initial Number	
		Even	Odd
Final Number	Even	d	a
	Odd	b	c

Table 1: Pattern a, b, c, d analysis

2.1.2. Divisibility Analysis

Since 892713821789371200 is divisible by 100 and 100 is divisible by 4, $N = 892713821789371200$ corresponds to term d.

2.1.3. Exponentiation Refinement

From $N = 892713821789371200$, we count each subsequent value, establishing $N = 1 \equiv a$, and $N = 2 \equiv b$. Hence, $N = 892713821789371238$ is equivalent to the $N = 38$th term, denoted as b. Thus, only the last two digits of N are required to find the repeating pattern S.

Examination of the 38th term:

$$38 \mod 4 = 2 \equiv \text{second term, hence the 38th term is b.}$$

Alternatively, for expedited evaluation:

2.1.4. Evenness Evaluation

Observing 38 as an even number.

2.1.5. Halving Procedure

Dividing 38 by 2, resulting in 19.

2.1.6. Oddness Recognition

Recognizing 19 as odd.

2.1.7. Divisibility Analysis

38 is even but not divisible by 4, implying divisibility by half of 4, which is 2. Thus, $38 \mod 4 = 2$, aligning with the 38th term being b.

Table 1 illustrates the determination of the initial number (last two digits) and the final number for the Nth term, based on whether the initial number is odd or even. For even initial numbers, the final number is obtained by dividing the initial number by 2. Conversely, for odd initial numbers, the final number is derived by subtracting 1 from the initial number and then dividing the result by 2.

2.2. Number Classification: Even and Odd

We initiate our analysis by classifying the given numbers into two primary categories: even and odd. Mathematically, this categorization is represented as:

- Even Numbers: $Ne = \{n \in \mathbb{Z} \mid n \text{ is even}\}$
- Odd Numbers: $No = \{n \in \mathbb{Z} \mid n \text{ is odd}\}$

2.3. Even Number Analysis

For even numbers $n \in Ne$, we employ an efficient analytical approach, using modular arithmetic principles. The algorithmic framework for identifying the corresponding term within the sequence S is as follows:

1.
   - Division by 2: Given an even number n, we perform the operation $k = n/2$, where k is an integer.
   - Even-Odd Classification: The parity of k determines the term's affiliation within the sequence S. Specifically, if k is even, then n corresponds to term "a" (the first term); otherwise, if k is odd, n aligns with term "b" (the second term).

This method capitalizes on the essential nature of even numbers, which always yield an integer quotient when divided by 2, creating an elegant association with the sequence terms.

2.4. Odd Number Analysis

Odd numbers $n \in No$ present a distinctive analytical challenge, necessitating a transformation prior to term identification. The algorithmic progression is as follows:

1. Subtraction and Division: Given an odd number n, we initiate by subtracting 1, yielding $m = n - 1$.
2. Division by 2: We subsequently perform the operation $k = m/2$, where k is an integer.
3. Even-Odd Classification: Analogous to the even number analysis, the parity of k delineates the term's correspondence within the sequence S. When k is even, the odd number n aligns with term "c" (the third term); alternatively, if k is odd, n corresponds to term "d" (the fourth term).

By this method, odd numbers traverse a mathematical journey encompassing subtraction and division, ultimately establishing a seamless association with the relevant sequence terms.

3. Discussion

In this section, we engage in an in-depth exploration of the empirical results presented in the "Results" section, unraveling the mathematical intricacies that underpin our analytical approach for determining terms in periodic sequences. We delve into the profound implications and applications of these findings within the realm of mathematical theory and computational sciences.

3.1. Application in Complex Numbers

3.1.1. Analysis of Powers of i with Application to $i^{892713821789371238}$

Consider the complex number i and its periodicity when subjected to integer exponentiation:

$$i^n \equiv \begin{cases} i & \text{if } n \equiv 1 \mod 4 \\ -1 & \text{if } n \equiv 2 \mod 4 \\ -i & \text{if } n \equiv 3 \mod 4 \\ 1 & \text{if } n \equiv 0 \mod 4 \end{cases}$$

This periodic behavior stems from the cyclical property of i, specifically $i^{4n} = 1$. Now, focusing on $i^{892713821789371238}$.

Proving that only the last two digits are necessary:

1. *Exponent Partitioning:* We commence by breaking down the exponent 892713821789371238 into two components: 892713821789371200 and 38.

2. *Divisibility Analysis:* The term 892713821789371200 has two zeros at the end. Therefore, it is divisible by 100, and since 100 is divisible by 4, it follows that 892713821789371200 is divisible by 4. Consequently, $i^{892713821789371200} = 1$.

		Initial Number	
		Even	Odd
Final Number	Even	1	i
	Odd	-1	-i

Table 2: Complex number Even/Odd analysis

3. *Exponentiation Refinement:* Hence, we establish $i^{892713821789371238} = 1 \cdot i^{38} = i^{38}$.

Hence, it can be proven that only the last two digits of the exponent are required to find the value of the complex number i.

Examination of i^{38}:

1. *Modular Interpretation:* Recognizing that $38 \mod 4 = 2$, we proceed to express i^{38} as $i^4 \cdot (9+2)$.

2. *Leveraging Cyclical Property:* We harness the periodic nature of i and its powers, resulting in

$$i^4 \cdot 9 = (i^4)^9 = 1^9 = 1$$

.

3. *Final Computation:* As a consequence, $i^{38} = 1 \cdot i^2 = -1$.

Alternatively, an expedited method for those less inclined to recognize the divisibility by 4 involves the following steps:

1. *Evenness Evaluation:* Noting that 38 is an even number.

2. *Halving Procedure:* Dividing 38 by 2, yielding $38/2 = 19$.

3. *Oddness Recognition:* Recognizing 19 as an odd number.

4. *Exponentiation Analysis:* Concluding that 38 is not divisible by 4 but is indeed even, implying divisibility by half of 4, which is 2. Consequently, the remainder of $38 \mod 4$ is 2.

Table 2 shows how the odd/even initial number (the last two digits) and final number (the number resulted when, if an even number is divided by 2 or, in the case of an odd number, it is subtracted by 1 and divided by 2) of the last two digits of the power of the complex number can be used to identify its value:

3.2. Conclusion

From the table above, it can be seen that if the initial number is even, the possible outcomes are 1 or -1. Similarly, if the initial number is odd, the possible outcomes are i and $-i$. In this way, an even initial number can be associated with a real number (1 or -1), while an odd initial number can be associated with an imaginary number (i or $-i$).

As for the final number, if the number is even, the possible outcomes are positive, which are either 1 or i. Similarly, if the final number is odd, the possible outcomes are negative, which are -1 or $-i$. In this way, an even final number can be associated with a positive number (i or 1), while an odd final number can be associated with a negative number ($-i$ or -1).

Hence, the initial number determines if the number is 1 or i, while the final number determines its sign, whether it is positive or negative.

The main purpose of this entire odd and even classification is to check the divisibility of the number by 4.

3.2.1. Computational Efficiency and Beyond

Beyond the realm of mathematical abstraction, our analytical methodologies manifest computational efficiency, offering a balance between mathematical insight and practical utility. The empirical confirmation of these methodologies across a diverse spectrum of numbers underscores their resilience in handling real-world scenarios. This computational efficiency is particularly pertinent in algorithm design, where precision and speed converge in a symbiotic relationship.

Understanding the cyclical behavior of powers of i and the associated congruence properties provides a valuable tool for math competitions, especially in scenarios involving complex number exponentiation. In competitions, time is of the essence, and being able to swiftly determine the value of i raised to a large exponent can save crucial seconds. By recognizing the periodicity of i powers modulo 4 and employing this understanding, competitors can quickly deduce the resulting term in the cycle, whether it's i, -1, $-i$, or 1. This approach significantly simplifies the computation of i raised to large powers, offering a strategic advantage and facilitating efficient solutions within the time constraints of competitive settings.

3.2.2. Applications in Cryptography and Number Theory

The analytical methodologies developed in this research paper bear direct implications for the fields of cryptography and number theory. Periodic sequences, often encountered in cryptographic algorithms, benefit from precise term identification. Our approach offers a systematic and efficient means to navigate these sequences, potentially enhancing cryptographic protocol design and decryption processes.

Within the domain of number theory, the analytical insights shown herein resonate with concepts of congruence, modular arithmetic, and residue classes. These findings enrich the mathematical arsenal employed in number theory, potentially advancing research in areas such as Diophantine equations and congruence properties of integers.

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5. Declaration of Interest

I declare that there is no conflict of interest related to this research paper. The work presented herein is the result of independent research, and no external organization or entity has influenced the content, analysis, or conclusions presented in this paper.

This research has been conducted with the primary goal of advancing knowledge and contributing to the field of mathematical analysis. The methodologies, findings, and interpretations presented in this paper are based on academic and scientific principles, and no personal or financial interests have influenced the research process or outcomes.

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